

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Product Description: **OPTIZYME™ 10X BUFFER 4**
Cat No. : **BP8087-1**

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

Details of the supplier of the safety data sheet

Importer

Fisher Scientific Korea
D5,D6, Incheon Airport Logistics Complex
150, Gonghangdong-Ro 296 Beon-Gil
Jung-Gu, Incheon
Tel: +82-1661-9555
Fax: +82-2-2023-0603

Supplier

Fisher Scientific
One Reagent Lane
Fair Lawn, NJ 07410
Tel: (201) 796-7100

E-mail address Chem.KR@thermofisher.com

Emergency Telephone Number

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970
CHEMTREC: 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887
Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

Based on available data, the classification criteria are not met

Environmental hazards

Based on available data, the classification criteria are not met

Label Elements

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Other Hazards

This product does not contain any known or suspected endocrine disruptors

NFPA

Health
1

Flammability
1

Instability
0

Physical hazards
N/A

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.2. Mixtures

Component	Common Name	CAS No	Index No	Weight %
Water	Aqua	7732-18-5	KE-35400	70 - 80
Potassium acetate	Potassium acetate	127-08-2	KE-29069	>=5 - <10
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	No information available	6850-28-8	Not listed	>=5 - <10
Magnesium acetate	No information available	142-72-3	KE-22674	2 - 2.5

SECTION 4: FIRST AID MEASURES

Description of first aid measures

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

Ingestion

Rinse mouth. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. Get medical attention.

Inhalation

Remove to fresh air. Get medical attention immediately if symptoms occur. If not breathing, give artificial respiration.

Self-Protection of the First Aider

No special precautions required.

Most important symptoms and effects, both acute and delayed

No information available.

Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

Extinguishing media

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Extinguishing media which must not be used for safety reasons

No information available.

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Special hazards arising from the substance or mixture

No information available.

Hazardous Combustion Products

None under normal use conditions.

Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation.

Environmental precautions

Do not flush into surface water or sanitary sewer system. Prevent further leakage or spillage if safe to do so. See Section 12 for additional Ecological Information.

Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Use spark-proof tools and explosion-proof equipment.

Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation.

Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Store product at -20C.

Specific End Uses

Use in laboratories.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Component	CAS No	Korea	ACGIH TLV	OSHA PEL
Water	7732-18-5	Not listed	Not listed	Not listed
Potassium acetate	127-08-2	Not listed	Not listed	Not listed
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not listed	Not listed	Not listed
Magnesium acetate	142-72-3	Not listed	Not listed	Not listed

Component	CAS No	European Union	The United Kingdom	Germany
Water	7732-18-5	Not listed	Not listed	Not listed
Potassium acetate	127-08-2	Not listed	Not listed	Not listed
1,3-Propanediol,	6850-28-8	Not listed	Not listed	Not listed

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2-amino-2-(hydroxymethyl)-, acetate (salt)				
Magnesium acetate	142-72-3	Not listed	Not listed	Not listed

ACGIH - Biological Exposure Indices

Component	CAS No	ACGIH - Biological Exposure Indices
Water	7732-18-5	Not listed
Potassium acetate	127-08-2	Not listed
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not listed
Magnesium acetate	142-72-3	Not listed

Exposure Controls

Engineering Measures

None under normal use conditions.

Personal protective equipment

Eye Protection

Wear safety glasses with side shields (or goggles)

Hand Protection

Protective gloves

Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Personal protective equipment

Use only those certified by the Korea Occupational Safety and Health Administration.

Respiratory Protection

No protective equipment is needed under normal use conditions

Recommended Filter type:

Particle filter

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

Environmental exposure controls

No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance (Physical State, Color, etc.) Colorless Liquid

Odor

Odorless

Odor Threshold

No data available

pH

7.5

Melting Point/Range

No data available

Softening Point

No data available

Boiling Point/Range

No information available

Flash Point

Not applicable

Method - No information available

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Evaporation Rate No data available
Flammability (solid,gas) Not applicable
Explosion Limits No data available

Liquid

Vapor Pressure No data available
Vapor Density No data available
Specific Gravity / Density 1.05
Bulk Density Not applicable
Water Solubility Miscible
Solubility in other solvents No information available

(Air = 1.0)

Liquid

Partition Coefficient (n-octanol/water)

Component	CAS No	log Pow
Water	7732-18-5	No data available
Potassium acetate	127-08-2	No data available
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	No data available
Magnesium acetate	142-72-3	No data available

Autoignition Temperature Not applicable
Decomposition Temperature No data available
Viscosity No data available
Explosive Properties No information available
Oxidizing Properties No information available

Molecular Formula Not applicable for mixtures
Molecular Weight Not applicable for mixtures

SECTION 10: STABILITY AND REACTIVITY

Reactivity No

Chemical Stability Stable under normal conditions.

Possibility of Hazardous Reactions

Hazardous Polymerization No information available.
Hazardous Reactions No information available.

Conditions to Avoid No information available.

Incompatible Materials No information available.

Hazardous Decomposition Products
None under normal use conditions.

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SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

Information on expected route of exposure

Inhalation	May cause irritation.
Ingestion	May cause irritation.
Eyes	May cause irritation.
Skin	May cause irritation.

Information on Health Hazards

(a) acute toxicity;

Oral	Based on available data, the classification criteria are not met
Dermal	No data available
Inhalation	No data available

Toxicology data for the components

Component	CAS No	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	7732-18-5	-	-	-
Potassium acetate	127-08-2	LD50 = 3250 mg/kg (Rat)	LD50 > 20000 mg/kg (Rabbit)	No data available
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	No data available	No data available	No data available
Magnesium acetate	142-72-3	LD50 = 8610 mg/kg (Rat)	LD50 > 20000 mg/kg (Rabbit)	No data available

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory	No data available
Skin	No data available

Component	CAS No	Test method	Test species	Study result
Water	7732-18-5	No data available	No data available	No data available
Potassium acetate	127-08-2	No data available	No data available	No data available
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	No data available	No data available	No data available
Magnesium acetate	142-72-3	No data available	No data available	No data available

No information available

(e) germ cell mutagenicity; No data available

Component	CAS No	Test method	Test species	Study result
Water	7732-18-5	No data available	No data available	No data available
Potassium acetate	127-08-2	No data available	No data available	No data available
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	No data available	No data available	No data available
Magnesium acetate	142-72-3	No data available	No data available	No data available

None known

(f) carcinogenicity; No data available

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Component	CAS No	Test method	Test species / Duration	Study result
Water	7732-18-5	No data available	No data available	No data available
Potassium acetate	127-08-2	No data available	No data available	No data available
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	No data available	No data available	No data available
Magnesium acetate	142-72-3	No data available	No data available	No data available

There are no known carcinogenic chemicals in this product

Component	CAS No	IARC	NTP	ACGIH	OSHA	UK
Water	7732-18-5	Not listed	Not listed	Not listed	Not listed	Not listed
Potassium acetate	127-08-2	Not listed	Not listed	Not listed	Not listed	Not listed
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not listed	Not listed	Not listed	Not listed	Not listed
Magnesium acetate	142-72-3	Not listed	Not listed	Not listed	Not listed	Not listed

(g) reproductive toxicity; No data available

Component	CAS No	Test method	Test species / Duration	Study result
Water	7732-18-5	No data available	No data available	No data available
Potassium acetate	127-08-2	No data available	No data available	No data available
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	No data available	No data available	No data available
Magnesium acetate	142-72-3	No data available	No data available	No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Other Adverse Effects

No information available.

Component	CAS No	EU - Endocrine Disruptors Candidate List	EU - Endocrine Disruptors - Evaluated Substances	Japan - Endocrine Disruptor Information
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Potassium acetate	127-08-2	Not applicable	Not applicable	Not applicable
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not applicable	Not applicable	Not applicable
Magnesium acetate	142-72-3	Not applicable	Not applicable	Not applicable

SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects

Component	CAS No	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
Water	7732-18-5	No data available	No data available	No data available	No data available
Potassium acetate	127-08-2	LC50: = 6800 mg/L, 96h semi-static (Oncorhynchus mykiss)	No data available	No data available	No data available

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1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	No data available	No data available	No data available	No data available
Magnesium acetate	142-72-3	No data available	No data available	No data available	No data available

Persistence and degradability

Persistence

Miscible with water, Persistence is unlikely, based on information available.

Bioaccumulative potential

Bioaccumulation is unlikely

Mobility in soil

The product is water soluble, and may spread in water systems . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

Ozone Depletion Potential

Component	CAS No	Ozone Depletion Potential
Water	7732-18-5	Not listed
Potassium acetate	127-08-2	Not listed
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not listed
Magnesium acetate	142-72-3	Not listed

Other adverse effects

No information available

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Waste from Residues/Unused Products

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

Contaminated Packaging

Empty containers should be taken to an approved waste handling site for recycling or disposal.

SECTION 14: TRANSPORT INFORMATION

Road and Rail Transport

Not Regulated

IATA

Not regulated

IMDG/IMO

Marine Pollutant

Not regulated

No hazards identified

Special Precautions for User

No special precautions required

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

International Inventories

Component	CAS No	KECL	TSCA	EINECS	IECSC	DSL	NDSL	PICCS	ENCS	ISHL	AICS
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Water	7732-18-5	KE-35400	X	231-791-2	X	X	-	X	X		X
Potassium acetate	127-08-2	KE-29069	X	204-822-2	X	X	-	X	X	X	X
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	-	X	229-939-6	-	-	X	X	-		-
Magnesium acetate	142-72-3	KE-22674	X	205-554-9	X	X	-	X	X	X	X

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable
Potassium acetate	127-08-2	Not applicable	Not applicable	Not applicable	Not applicable
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not applicable	Not applicable	Not applicable	Not applicable
Magnesium acetate	142-72-3	Not applicable	Not applicable	Not applicable	Not applicable

Component	CAS No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential
Water	7732-18-5	Listed	Not applicable	Not applicable
Potassium acetate	127-08-2	Listed	Not applicable	Not applicable
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not applicable	Not applicable	Not applicable
Magnesium acetate	142-72-3	Listed	Not applicable	Not applicable

Korean National Regulations

Component	CAS No	Act on Registration and Evaluation of Chemical Substances (K-REACH)	Authorised Chemicals	Existing Substances Subject to Registration
Water	7732-18-5	Annex 1 - KE-35400 Exempt (Index No. 25)	Not applicable	Not applicable
Potassium acetate	127-08-2	Annex 1 - KE-29069	Not applicable	Not applicable
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not applicable	Not applicable	Not applicable
Magnesium acetate	142-72-3	Annex 1 - KE-22674	Not applicable	Not applicable

Component	CAS No	Chemical Control Act - Toxic Chemicals	Chemical Control Act - Prohibited Chemicals	Chemical Control Act - Use Restricted Chemicals
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Potassium acetate	127-08-2	Not applicable	Not applicable	Not applicable
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not applicable	Not applicable	Not applicable
Magnesium acetate	142-72-3	Not applicable	Not applicable	Not applicable

Component	CAS No	Chemical Control Act - Accident Precaution Chemicals (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures)
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Potassium acetate	127-08-2	Not applicable	Not applicable	Not applicable
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not applicable	Not applicable	Not applicable
Magnesium acetate	142-72-3	Not applicable	Not applicable	Not applicable

Component	CAS No	Waste Control Law	Ministry of Environment - CMR risk	Ministry of Environment - Critically Controlled Substance

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Water	7732-18-5	Not applicable	Not applicable	Not applicable
Potassium acetate	127-08-2	Not applicable	Not applicable	Not applicable
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not applicable	Not applicable	Not applicable
Magnesium acetate	142-72-3	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Harmful Agents Subject to Work Environment Monitoring	ISHA - Prohibited substances	ISHA - Substances requiring permission
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Potassium acetate	127-08-2	Not applicable	Not applicable	Not applicable
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not applicable	Not applicable	Not applicable
Magnesium acetate	142-72-3	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Substances subject to control	ISHA - Harmful Agents Requiring Health Examination	ISHA - Permissible Exposure Limits
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Potassium acetate	127-08-2	Not applicable	Not applicable	Not applicable
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not applicable	Not applicable	Not applicable
Magnesium acetate	142-72-3	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Subject to Process Safety Reports (minimum quantity)	ISHA - Threshold Limit Values (TLVs) Chemicals	ISHA - Special management materials
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Potassium acetate	127-08-2	Not applicable	Not applicable	Not applicable
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not applicable	Not applicable	Not applicable
Magnesium acetate	142-72-3	Not applicable	Not applicable	Not applicable

National Fire Association - Dangerous Substances Minimum quantity requiring a permit

Component	CAS No	Class 1 - Oxidising solids	Class 2 - Flammable solid	Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet	Class 4 - Flammable liquids	Class 5 - Self-reactive substances	Class 6 - Oxidising liquids
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Potassium acetate	127-08-2	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
1,3-Propanediol, 2-amino-2-(hydroxymeth yl)-, acetate (salt)	6850-28-8	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable
Magnesium acetate	142-72-3	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable

Control Parameters

Component	CAS No	Korea	ACGIH - Biological Exposure Indices
Water	7732-18-5	Not listed	Not listed
Potassium acetate	127-08-2	Not listed	Not listed
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not listed	Not listed
Magnesium acetate	142-72-3	Not listed	Not listed

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US Management Information

OSHA - Occupational Safety and Health Administration
Not applicable

Component	CAS No	Specifically Regulated Chemicals	Highly Hazardous Chemicals
Water	7732-18-5	Not applicable	Not applicable
Potassium acetate	127-08-2	Not applicable	Not applicable
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not applicable	Not applicable
Magnesium acetate	142-72-3	Not applicable	Not applicable

CERCLA Not applicable

Component	CAS No	CERCLA Extremely Hazardous Substances RQs	Hazardous Substances RQs	SARA 313 - Threshold Values %
Water	7732-18-5	Not applicable	Not applicable	Not applicable
Potassium acetate	127-08-2	Not applicable	Not applicable	Not applicable
1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, acetate (salt)	6850-28-8	Not applicable	Not applicable	Not applicable
Magnesium acetate	142-72-3	Not applicable	Not applicable	Not applicable

GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Based on available data, the classification criteria are not met.

SECTION 16: OTHER INFORMATION

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

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Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Creation Date	16-Sep-2011
Revision Date	06-Jun-2024
Revision Number	5
Revision Summary	SDS sections updated.

MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet